Mixed Noise-Oriented Hyperspectral and Multispectral Image Fusion

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***Abstract*— Hyperspectral images (HSIs) possess the capability to accurately characterize the attribute information of objects. However, they are usually obtained at a high spectral resolution with a compromise of their spatial resolution. In addition, they are easily contaminated by mixed noise induced by instrument and atmospheric effects. These disadvantages, to a certain degree, hinder the interpretations and applications of the HSIs. To overcome these limitations, in this article, we propose a novel mixed noise-oriented hyperspectral and multispectral image fusion method, termed (MixFus). First, a sparse noise detection method is proposed by leveraging a subset of specifically chosen hyperspectral bands to estimate noise in HSIs and then employing Gaussian mixture models (GMMs) to detect sparse noise from the estimated noise. Then, a robust subspace estimation method is introduced by replacing the detected sparse noise with new estimates using median values within a sliding window for a better estimation of the subspace, which offers improved accuracy and robustness of subspace estimation. Finally, in addition to the introduction of a state-of-the-art image prior based on the plug-and-play technique to exploit self-similarity characteristics in the eigen-images, we also impose a weighted group sparse regularization on the eigen-images to better promote the group sparsity of the spatial differences between the eigen-images, which further improve the denoising performance. We evaluate the proposed method by performing extensive experiments on three reduced-resolution HSIs and a full-resolution HSI in comparison with seven state-of-the-art competitors. Experimental results demonstrate the superiority of the proposed method over the competitors in the fusion of hyperspectral and multispectral images against mixed noise.**

***Index Terms*— Hyperspectral images (HSIs), image fusion, mixed noise, multispectral images, super-resolution.**

NOMENCLATURE

Notations	Interpretations
$\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$	3-D tensor.
$\mathbf{X}$	Matrix.
$\mathbf{x}$	Vector.
x	Scalar.
order N	Number of dimensions in a tensor.
mode n	nth dimension of a tensor.
$\mathcal{X}(:,:,i)$	ith mode-3 slice of $\mathcal{X}$.
$\mathbf{X}_{(3)} \in \mathbb{R}^{I_3 \times (I_1 * I_2)}$	Mode-3 unfolding of $\mathcal{X}$.
$\|\mathcal{X}\|_F$	Frobenius norm of tensor $\mathcal{X}$: $\|\mathcal{X}\|_F = \sqrt{\sum_{i_1,\ldots,i_N} \|x_{i_1,\ldots,i_N}\|^2}$.
$\|\mathcal{X}\|_{2,1}$	$\ell_{2,1}$ norm of tensor $\mathcal{X}$: $\|\mathcal{X}\|_{2,1} = \sum_{i_1,i_2} \|x_{i_1,i_2,:}\|_2$.
$\odot$	Componentwise multiplication.
$\times_k$	Mode-k tensor–matrix product.

I. INTRODUCTION

HYPERSPECTRAL imaging technology can capture both spatial and spectral information of observed targets through simultaneous acquisition of images and continuous spectra within the visible to near-infrared wavelengths [1]. The benefits of hyperspectral images (HSIs) have become increasingly evident, leading to the launch of numerous spectrum imaging research programs worldwide. With an increasing number of imaging spectrometers available, hyperspectral imaging is now widely employed across a range of sectors, including anomaly detection [2], geological surveying, and medical research [3].

However, due to the limitations of the imaging platform, an HSI is usually obtained at a high spectral resolution with a compromise of its spatial resolution, which, to a certain degree, limits the application of HSIs. Therefore, improving the spatial resolution from the observed low-resolution HSI (LR-HSI) is an important preprocessing task before further applications [4]. Generally, the resolution reconstruction of HSIs can be broadly categorized into two types: super-resolution using a single LR-HSI and fusion reconstruction using an LR-HSI and a high-resolution multispectral image (HR-MSI). Since image reconstruction is an ill-posed problem, it is extremely challenging to recover a high-resolution HSI (HR-HSI) using a single data source. To facilitate the reconstruction process and make it more meaningful, HR-MSIs are introduced for the fusion, serving as suitable auxiliary data. Thence, developing high-performance algorithms for effectively fusing the spatial and spectral information of LR-HSI and HR-MSI to improve the spatial resolution of HSIs has become a popular research field.

Apart from the low spatial resolution of HSIs, in practice, noise commonly exists in some widely used HSIs, such as

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GF-5 [5], EO-1 [6], HYDICE [7], and more, which cannot be ignored in practical applications. Since the bandwidth of the electromagnetic wave scattered into the instantaneous field of view is narrow, the increase of spectral bands makes fewer photons that can be received by each band, leading to the proneness of the HSI to noise contamination. The noise can be induced by the imaging device (charge-coupled device) during the conversation from electrons to the final image [6]. Apart from instrument noise, noise can also be introduced by atmospheric effects during the acquisition of the images and optical aberrations and misalignments in imaging systems [8]. Thus, noise in the HSIs can be a complex mixture, including common noise such as Gaussian noise and Poissonian noise, as well as sparse noise such as "Salt & Pepper" noise and deadlines/stripes [9]. The complex noise introduced not only severely affects the visualization of HSIs, but also severely degrades the quality of HSIs, and finally affects the accuracy of downstream applications. Therefore, in many practical applications, noise removal has become a critical factor to be considered for improving data quality and ensuring the accuracy of specific applications, such as classification [10], water monitoring [11], and anomaly detection [12], [13]. As a result, HSI denoising is of great significance and worthy of in-depth study, attracting increasing interest year by year since 2001 [5].

Since the acquisition of HR-MSI is popular, we can easily acquire a noise-free HR-MSI compared with HSIs. Thus, it is very feasible and meaningful to improve the resolution of a noise-degraded HSI with the aid of a noise-free HR-MSI. A straightforward way to obtain the noise-free fused image is by performing denoising on the fused image obtained by existing fusion methods or denoising the HSIs before image fusion. However, although denoising before or after image fusion can remove the noise to some degree, it can also introduce new errors or fail to remove all the noise. In addition, the separating operations make the parameter selection very hard and the procedure to be troublesome. Consequently, a noise-robust hyperspectral and multispectral image fusion method, which has rarely been investigated, is needed to improve spatial resolution while simultaneously restoring HSI degradations.

In this article, considering the common existence of mixed noise on the observed LR-HSIs, we propose a novel mixed noise-oriented hyperspectral and multispectral image fusion method, termed (MixFus) by estimating the subspace robustly and regularizing the eigen-images effectively. The proposed method aims to reconstruct an HR-HSI from an LR-HSI contaminated by mixed noise (such as Gaussian noise, "Salt & Pepper" noise, deadlines, or stripes) and an HR-MSI containing only simple Gaussian noise. Contributions of this article are given in detail in the following.

1) We propose a sparse noise detection method by first leveraging a subset of specifically chosen hyperspectral bands to estimate noise in HSIs and then employing Gaussian mixture models (GMMs) to detect sparse noise from the estimated noise, which allows us to mitigate the influence of the sparse noise on the fusion performance.
2) A robust subspace estimation method is introduced. Before the subspace estimation, the detected sparse noise is replaced with new estimates using median values within a sliding window for a better estimation of the subspace, which offers improved accuracy and robustness of subspace estimation.
3) We combine a state-of-the-art image prior and a weighted group sparse regularization to fully exploit the self-similarity characteristics in the eigen-images and the group sparsity of the spatial differencing of the eigen-images along the mode-3, respectively, which further improves the fusion performance of the proposed method.

Extensive experiments on three reduced-resolution images with different types of noise and a full-resolution image are carried out to fully evaluate the fusion performance of the proposed method in comparison with seven state-of-the-art fusion competitors. We organize the remainder of this article as follows. The related work of this article is introduced in Section II. In Section III, we introduce the proposed mixed noise-oriented fusion method in detail. In Section IV, we thoroughly evaluate and analyze the fusion performance of the proposed method in comparison with seven state-of-the-art fusion methods using three reduced-resolution experiments and one full-resolution experiment. Last but not least, we conclude this article in Section V.

II. Related Work

The purpose of fusing multispectral and HSIs is to fully leverage the spectral information available in LR-HSI and the spatial information provided by HR-MSI to generate an HSI with high spatial resolution. The challenge of utilizing the traits of MSIs and HSIs for effective image fusion has garnered significant research interest. Based on their attributes, existing fusion techniques can be classified into four broad categories: pansharpening-based methods, matrix decomposition-based methods, tensor decomposition-based methods, and deep-learning-based methods.

Generally, image fusion involves the fusion of panchromatic with multispectral or panchromatic with HSIs, commonly referred to as pansharpening techniques [14], [15], [16]. These techniques can be adapted to the fusion of HSIs and MSIs after some changes. For instance, one method involves dividing the channels of an HSI into regions and then applying the pansharpening technique to fuse the corresponding regions of the HSIs and MSIs [17]. Another approach for enhancing the spatial texture structure of each hyperspectral band involves weighting and combining all bands of the MSI, thereby obtaining high-resolution spatial information [18], such as GLPHS [19].

High spatial and spectral resolution images can be acquired using matrix decomposition. Matrix decomposition-based methods can be broadly classified into sparse representation methods and low-rank representation-based methods. The sparse representation methods regard the spectral basis as the over-complete dictionary which can be learned via sparse dictionary learning algorithms, such as K-SVD [48], and

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online dictionary learning [49]. Low-rank representation-based methods fuse the images by exploiting the inherent spectral characteristics of the scene via a subspace spanned by a set of spectral signatures. For example, Yokoya et al. [20] proposed a coupled nonnegative matrix factorization (CNMF) method, which estimates the endmember and abundance matrices via alternating spectral unmixing. Simões et al. [21] proposed a method called HySure based on the vector total variation to regularize the spatial distribution of subspace coefficients, where the subspace can be defined either by singular value decomposition (SVD) [22] or vertex component analysis (VCA) [23]. Recently, growing attention has been dedicated to tackling the HSI and MSI fusion with spectral variability phenomenon, such as FuVar [24] and HMF-IPNMF [25]. Except for the above-mentioned methods, Kwan et al. [26] first applied the block-matching and 3-D (BM3D) filtering on LR-HSI to achieve super-resolution reconstruction and then utilized a matrix-based hybrid color mapping technique with high-resolution color images to generate a high-quality fused image.

Multispectral and hyperspectral data can be represented by a 3-D cube, which can naturally be depicted as a third-order tensor. Tensor decomposition-based techniques can effectively analyze features in the spectral and spatial dimensions. Tucker decomposition [27] is a commonly used tensor decomposition method. For example, Li et al. [59] proposed a coupled sparse tensor factorization (CSTF) method in which the estimation of the dictionaries and the core tensor are formulated as a coupled tensor factorization of the LR-HSI and the HR-MSI. Dian et al. [28] propose a nonlocal sparse tensor factorization (NLSTF) approach for fusing MSIs and HSIs. Canonical polyadic (CP) decomposition [29] is a special case of Tucker decomposition. Xu et al. [30] proposed a nonlocal coupled tensor CP decomposition method (NCTCP) for fusing MSIs with HSIs by assuming that LR-HSIs and HR-MSIs have the same factor matrices in CP decomposition. Additionally, other tensor-based methods such as tensor SVD (t-SVD)-based methods [31], tensor train decomposition-based methods [32], tensor ring decomposition-based methods [33], and low-rank tensor-approximation (LRTA)-based [34] methods have also been proposed to fusion HSIs and MSIs.

Over the past few years, deep learning has demonstrated vast potential in image processing, and as a result, has opened up a new avenue to address the challenge of fusing HSIs and MSIs. For example, Xie et al. [35] learned the image prior from the deep-learning approach. Convolutional neural networks are employed to handle the fusion of hyperspectral and multispectral images [36], [37], enabling better extraction of detailed information. Recently, Transformer-based methods [38], followed by a multiscale version [39], have been proposed, achieving excellent performance. Qu et al. [40] and [41] introduced an unsupervised sparse Dirichlet network, which adopts an unsupervised encoder–decoder architecture to extract the spatial information and spectral information of two modalities. Deep-learning-enabled techniques attain the fusion of HSIs and MSIs even in the presence of unknown degradation models and infer the inherent features of images effectively. Nonetheless, they necessitate a large amount of data for training, which is often inaccessible in realistic scenarios. Furthermore, deep neural networks exhibit weak generalization ability, as the spectroscopic resolution, band number, and space resolution of fused data may vary considerably, rendering a neural network model developed for a specific dataset sometimes inadequate for other datasets [42].

III. Proposed MixFus Method

In this section, we will introduce the proposed fusion method in detail. Following the nomenclatures of [43], we first give the notations and their interpretations used in this article in Nomenclature. The flowchart of the proposed fusion method is illustrated in Fig. 1.

A. Observation Model

Given the LR-HSI, $\mathcal{X} \in \mathbb{R}^{w\times h\times L}$, and the HR-MSI, $\mathcal{Y} \in \mathbb{R}^{W\times H\times l}$, where w, h, and L are the number of rows, columns, and bands in LR-HSIs, respectively, W, H, and L are the number of rows, columns, and bands in HR-MSIs, respectively. Let $\mathcal{Z} \in \mathbb{R}^{W\times H\times L}$ denote the underlying HR-HSI. The mode-3 unfolding of the HR-HSI, LR-HSI, and HR-MSI can be expressed as $\mathbf{Z}_{(3)} \in \mathbb{R}^{L\times WH}$, $\mathbf{X}_{(3)} \in \mathbb{R}^{L\times wh}$, $\mathbf{Y}_{(3)} \in \mathbb{R}^{l\times WH}$, respectively.

The observed LR-HSI, $\mathcal{X}$, can be viewed as the spatially degraded version of the HR-HSI, $\mathcal{Z}$, and is written as

$$\mathbf{X}_{(3)} = \mathbf{Z}_{(3)}\mathbf{B}\mathbf{S} + \mathbf{N}_x \tag{1}$$

where $\mathbf{B} \in \mathbb{R}^{WH\times WH}$ is a blur kernel that encapsulates the hyperspectral sensor's spatial point-spread function (PSF) describing the sensor's spatial degradation process, $\mathbf{S} \in \mathbb{R}^{WH\times wh}$ is the downsampling matrix, and $\mathbf{N}_x \in \mathbb{R}^{L\times wh}$ is additive noise matrix, which may include Gaussian noise, "Salt & Pepper" noise, deadlines, or stripes.

The HR-MSI, $\mathcal{Y}$, can be viewed as the spectral degraded version of $\mathcal{Z}$ and can be represented as

$$\mathbf{Y}_{(3)} = \mathbf{R}\mathbf{Z}_{(3)} + \mathbf{N}_y \tag{2}$$

where $\mathbf{R} \in \mathbb{R}^{l\times L}$ is the spectral response function and $\mathbf{N}_y \in \mathbb{R}^{l\times WH}$ is the Gaussian noise matrix. The spectral bands of HSIs have a high correlation, which means that spectral vectors (i.e., mode-3 vectors) of an HSI can be well approximated by vectors from a low-dimensional subspace [44], [45], [46].

Based on the subspace representation, the HR-HSI can be expressed as

$$\mathcal{Z} = \mathcal{A}\times_3\mathbf{E} + \mathcal{O} \tag{3}$$

where $\mathbf{E} \in \mathbb{R}^{L\times p}$ ($p \ll L$) is an orthogonal subspace basis for HSIs. $\mathcal{A} \in \mathbb{R}^{W\times H\times p}$ is a representation tensor whose entries are coefficients correspond to $\mathbf{E}$. $\mathcal{O} \in \mathbb{R}^{W\times H\times L}$ is a residual tensor that represents the subtle differences between the subspace representation of an image and its original counterpart $\mathcal{Z}$. The mode-3 slices of $\mathcal{Z}$ are also called eigenimages. In addition, the subtle residual represented by $\mathcal{O}$ can be neglected in low-dimensional subspace representation.

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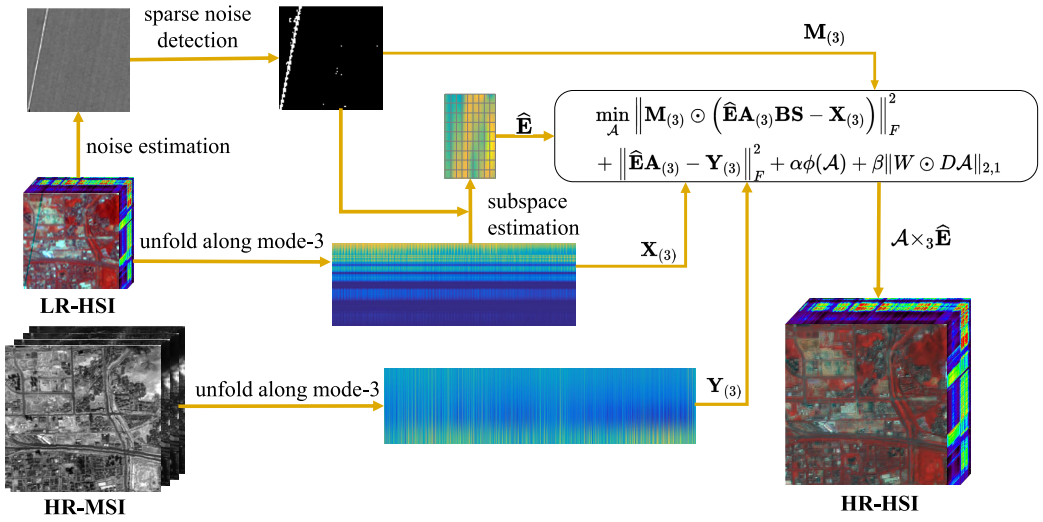


Fig. 1. Flowchart of the proposed fusion method.

Therefore, the equation can be approximately equivalent to the following representation:

$$\mathbf{Z}_{(3)} = \mathbf{E}\mathbf{A}_{(3)}. \tag{4}$$

B. Noise Estimation and Sparse Noise Detection

1) Noise Estimation: Since the LR-HSI retains the most spectral information of the HR-HSI, the subspace can be estimated from the LR-HSI using subspace learning methods, such as SVD. However, SVD will not work when the LR-HSI is contaminated by a mixture of Gaussian and sparse noise. Thus, it is necessary to detect and remove sparse noise before subspace estimation. To this aim, we first estimate the underlying mixture of noise and subsequently detect the sparse noise components within the mixture of noise. There exists a high degree of correlation among the spectral bands in HSIs, which suggests that any given band in HSIs can be approximated by a linear combination of the remaining $L-1$ bands. The process of the linear combination can be expressed as

$$\left[\mathbf{X}_{(3)}\right]_{i,:} = \gamma_i \left[\mathbf{X}_{(3)}\right]_{\varphi_i} + \boldsymbol{\xi}_i \tag{5}$$

where $[\cdot]_{i,:}$ represents the ith row element in the matrix and $[\cdot]_{\varphi_i}$ represents the new matrix formed after removing the ith row from the matrix. $\gamma_i \in \mathbb{R}^{1\times(L-1)}$ means regression coefficients. $\boldsymbol{\xi}_i \in \mathbb{R}^{1\times wh}$ means regression error and is considered the noise in the ith band.

However, the presence of numerous spectral bands in HSIs can result in spectral redundancy. Furthermore, noise that arises from sensor instruments and atmospheric propagation can degrade the quality of certain bands of the HSI. Utilizing polluted spectral bands for linear fitting operations can significantly affect the accuracy of the final noise estimation. Hence, the optimal neighborhood reconstruction (ONR) [47] method is first employed to conduct band selection for HSIs. This method is based on the correlation between adjacent bands in HSIs and involves the application of a noise reducer to minimize the impact of noisy bands, thereby selecting bands that are purer for spectral linear fitting. The band selection can be expressed as

$$\mathbf{X}_{\text{select}} = \text{ONR}(\mathbf{X}_{(3)}) \tag{6}$$

where $\mathbf{X}_{\text{select}} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k]^T$ is a matrix of selected bands and $k \leqslant L$. Then, the linear fitting process can be expressed as

$$\left[\mathbf{X}_{(3)}\right]_{i,:} = \gamma_i \mathbf{X}_{\text{select}} + \boldsymbol{\xi}_i. \tag{7}$$

$\mathbf{x}_k$ will be removed from $\mathbf{X}_{\text{select}}$ when$[\mathbf{X}_{(3)}]_{i,:} = \mathbf{x}_k$. The regression coefficients can be obtained using the least-squares method, that is,

$$\begin{aligned}\hat{\gamma}_i &= \arg\min_{\gamma_i} \left\|\left[\mathbf{X}_{(3)}\right]_{i,:} - \gamma_i \mathbf{X}_{\text{select}}\right\|_F^2 \\ &= \left[\mathbf{X}_{(3)}\right]_{i,:} \mathbf{X}_{\text{select}}^T \left(\mathbf{X}_{\text{select}} \mathbf{X}_{\text{select}}^T\right)^{-1}.\end{aligned} \tag{8}$$

Next, the noise in the ith band $\hat{\boldsymbol{\xi}}_i$ can be estimated by

$$\hat{\boldsymbol{\xi}}_i = \left[\mathbf{X}_{(3)}\right]_{i,:} - \hat{\gamma}_i \mathbf{X}_{\text{select}}. \tag{9}$$

2) Sparse Noise Detection: Generally speaking, if the ith band of LR-HSIs is contaminated by only Gaussian noise, $\hat{\boldsymbol{\xi}}_i$ can be approximated by a Gaussian distribution. However, in the presence of sparse noise, such as stripes and "Salt & Pepper" noise, a Gaussian distribution would not adequately model the noise well. Therefore, the GMM is employed to model the noise. Among these distributions, the one with the largest proportion of components is regarded as Gaussian noise, while the remaining noise, which does not conform to the Gaussian distribution, can be identified as sparse noise.

Specifically, a normality test is performed on $\hat{\boldsymbol{\xi}}_i$ using the kurtosis and skewness. The thresholds of kurtosis and skewness are set to 5 and 0.5, empirically. If the kurtosis and the skewness are smaller than the predetermined threshold, $\hat{\boldsymbol{\xi}}_i$ is considered to conform to the Gaussian distribution. Conversely, $\hat{\boldsymbol{\xi}}_i$ is deemed to include mixed noise. In this context, to simplify the formula notation, the band index of $\hat{\boldsymbol{\xi}}_i = [\hat{\xi}_{i,1}, \hat{\xi}_{i,2}, \ldots, \hat{\xi}_{i,wh}]^T$ will be omitted and it is written

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as $\hat{\boldsymbol{\xi}} = [\hat{\xi}_1, \hat{\xi}_2, \ldots, \hat{\xi}_{wh}]^T$. A GMM with three components is utilized to model mixed noise in the following form:

$$f\left(\hat{\xi}_j, \boldsymbol{\theta}\right) = \sum_{k=1}^{3} \pi_k \mathcal{N}\left(\hat{\xi}_j; \mu_k, \sigma_k^2\right) \tag{10}$$

where π_k is the generation probability of the kth component, and it is required to be nonnegative and sum to one. $\mathcal{N}(\hat{\xi}_j; \mu_k, \sigma_k^2)$ is the Gaussian probability density function with mean μ_k and variance σ_k^2. $\boldsymbol{\theta} = (\pi_1, \pi_2, \pi_3, \mu_1, \mu_2, \mu_3, \sigma_1^2, \sigma_2^2, \sigma_3^2)^T$ means a vector of parameters to be solved. The maximum likelihood estimate of $\boldsymbol{\theta}$ is written as

$$\begin{aligned}\widehat{\boldsymbol{\theta}} &= \arg\max_{\boldsymbol{\theta}} \log \mathcal{L}(\boldsymbol{\theta}) \\ &= \arg\max_{\boldsymbol{\theta}} \sum_{j=1}^{wh} \log\left\{\sum_{k=1}^{3} \pi_k \mathcal{N}\left(\hat{\xi}_j; \mu_k, \sigma_k^2\right)\right\}.\end{aligned} \tag{11}$$

The maximum likelihood problem stated above can be effectively tackled using the expectation maximization (EM) algorithm, thus deriving the optimal parameter set $\widehat{\boldsymbol{\theta}}$. The posterior probability of element $\hat{\xi}_j$ belonging to component k can be determined through the fit posterior probability. The component label $L_j \in \{1, 2, 3\}$ for each $\hat{\xi}_j$ can be obtained as

$$L_j = \arg\max_k \frac{\hat{\pi}_k \mathcal{N}\left(\hat{\xi}_j; \hat{\mu}_k, \hat{\sigma}_k^2\right)}{f\left(\hat{\xi}_j, \widehat{\boldsymbol{\theta}}\right)}. \tag{12}$$

Finally, Gaussian noise is identified by counting the number of component labels. The elements of $\hat{\boldsymbol{\xi}}_i$ corresponding to the label with the most occurrences are identified as Gaussian noise, while the rest are identified as sparse noise. A sparse noise mask $\mathcal{M} \in \mathbb{R}^{w \times h \times L}$ can be obtained by setting all samples detected as Gaussian noise to 1 and the remaining samples to 0.

C. Subspace Estimation

Normally, subspace can be estimated from the LR-HSI using subspace learning methods like SVD, as the LR-HSI retains the most spectral information of the HR-HSI. Mathematically, subspace $\mathbf{E} = \mathbf{U}(:, 1 : p)$, where $\{\mathbf{U}, \boldsymbol{\Sigma}, \mathbf{V}\} = \text{SVD}(\mathbf{X}_{(3)})$, $\mathbf{U} \in \mathbb{R}^{L \times L}$, and $\mathbf{V} \in \mathbb{R}^{p \times p}$ are orthogonal matrices, $\boldsymbol{\Sigma}$ is a diagonal matrix whose diagonal elements are singular values ranked by nonincreasing magnitude.

However, for the fusion and restoration of HSIs, the difficulties of utilizing subspace representation lie in the accurate estimation of the subspace from observations contaminated by not only Gaussian noise, but also sparse noise, and the effective regularization of representation coefficient $\mathcal{A}$ by fully exploiting the information of the eigen-images. These challenges motivate us to propose a subspace representation-based fusion method by estimating subspace robustly from the noisy LR-HSI and regularizing the eigen-images effectively. To estimate subspace robustly from the noisy LR-HSI, an effective subspace estimation method is introduced after the sparse noise mask is obtained.

Let $\mathbf{x}_j = [x_1, x_2, \ldots, x_{n_b}]^T \in \mathbb{R}^{n_b}$ denote the jth column vector of the observed HSI matrix $\mathbf{X}_{(3)}$, and q denote the length of a sliding window. The median for a sample x_i within a sliding window of q can be written as

$$\begin{aligned}m_i = \text{median}\big(&x_{i-q}, x_{i-q+1}, x_{i-q+2}, \ldots, x_i, \ldots, \\ &x_{i+q-2}, x_{i+q-1}, x_{i+q}\big)\end{aligned} \tag{13}$$

where median$(\cdot)$ is the function that returns the median values. In Section III-B, the matrix of sparse noise mask $\mathbf{M}_{(3)}$ has been obtained. These samples containing sparse noise can be regarded as outliers. If x_i is declared as an outlier, it is replaced with m_i as

$$\tilde{x}_i = \begin{cases} m_i, & \text{if } [\mathbf{M}_{(3)}]_{j,i} = 0 \\ x_i, & \text{otherwise.} \end{cases} \tag{14}$$

The new estimates may not be the exact spectral values of the clean HSIs. However, by replacing the detected samples in the observed image with a new estimate closer to the truth, we can obtain a new image without severe sparse noise, which will greatly benefit the estimation of subspace basis.

After replacing all the detected samples, we can obtain the new image $\widetilde{\mathcal{X}}$. Then, we can easily estimate the subspace from $\widetilde{\mathbf{X}}$, which contains approximately Gaussian noise. Specifically, the subspace can be obtained by performing SVD on $\widetilde{\mathbf{X}}$ and extracting the first p left-singular vectors

$$\widehat{\mathbf{E}} = \widetilde{\mathbf{U}}(:, 1 : p) \tag{15}$$

where $\{\widetilde{\mathbf{U}}, \widetilde{\boldsymbol{\Sigma}}, \widetilde{\mathbf{V}}\} = \text{SVD}(\widetilde{\mathbf{X}}_{(3)})$, $\widetilde{\mathbf{U}} \in \mathbb{R}^{L \times L}$, and $\widetilde{\mathbf{V}} \in \mathbb{R}^{p \times p}$ are orthogonal matrices, and $\widetilde{\boldsymbol{\Sigma}}$ is a diagonal matrix. The dimension of subspace, p, can be automatically estimated using the HySime method [48].

D. Characteristics in Eigen-Images

Like natural images, HSIs are images of the real world, thus they generally contain many similar patches at different locations and scales, which is called self-similarity. Since the bands of an HSI are self-similar and each band has the same pattern of self-similarity, then its corresponding eigen-images also have the property of self-similarity [49]. The exploitation of self-similarity in the eigen-images achieves state-of-the-art denoising performance, such as FastHyDe [49] and NG-meet [50].

In addition, since different bands present the same scene of the ground, different bands in HSIs should share the same patterns of piecewise smooth structure. In other words, the differencing images at two spatial directions across the entire bands should have group sparse characteristics along the spectral dimension. We use a subimage of Washington DC Mall data as an example to illustrate the group sparse characteristics of the spatial differences images. The spatial horizontal and vertical differencing images are shown in Fig. 2(a) and (b), respectively. In Fig. 2(a) and (b), the group sparsity pattern along the spectral mode can be clearly observed. We present the histogram of the ℓ_2-norm value of all spectral vectors of Fig. 2(a) and (b) in Fig. 2(c) and (d). It can be clearly seen from Fig. 2(c) and (d) that most of the ℓ_2-norm value tends to be zero, indicating the sparsity of the ℓ_2-norm value. Since eigen-images of the HSI share the same spatial patterns with the original HSI, it can be easily known that the spatial

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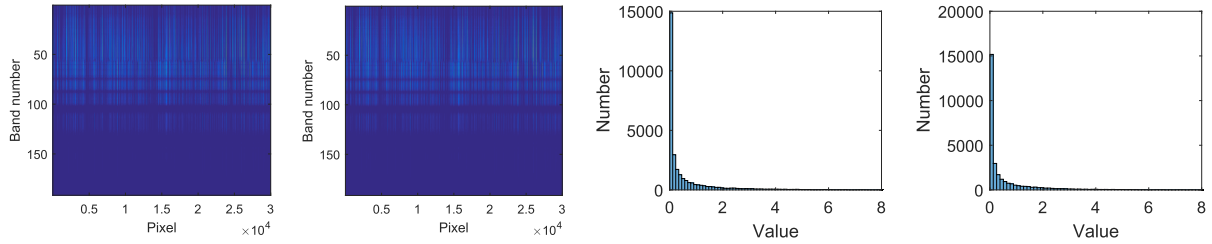


Fig. 2. Group sparse characteristics of the spatial differences images in a subimage of Washington DC Mall data. (a) Spatial horizontal difference image. (b) Spatial vertical difference image. (c) Histogram of (a). (d) Histogram of (b).

difference images of the eigen-images also have group sparsity along the third mode.

Thus, in this article, except for the nonlocal before exploiting self-similarity of the eigen-images, we introduce a weighted group sparse regularization [51] to promote the group sparsity of the spatial difference images along the third mode in the eigen-images as

$$\mathcal{J}(\mathcal{Z}) = ||W \odot D\mathcal{A}||_{2,1} \tag{16}$$

where D is the two differential operators in two spatial dimensions D_x and D_y, and W is the weight operator in two spatial dimensions. The two differential operators are calculated as follows:

$$\begin{cases} D_x\mathcal{A}(i,j,k) = \mathcal{A}(i,j+1,k) - \mathcal{A}(i,j,k) \\ D_y\mathcal{A}(i,j,k) = \mathcal{A}(i+1,j,k) - \mathcal{A}(i,j,k). \end{cases} \tag{17}$$

The weighted group sparse regularization is formulated as

$$\|W \odot D\mathcal{A}\|_{2,1} = \sum_{i=1}^{m}\sum_{j=1}^{n} W_x(i,j,:)\|D_x\mathcal{A}(i,j,:)\|_2 + \sum_{i=1}^{m}\sum_{j=1}^{n} W_y(i,j,:)\left\|D_y\mathcal{A}(i,j,:)\right\|_2. \tag{18}$$

E. Proposed Model and Optimization

In this article, to mitigate the impact of sparse noise on the fusion performance, first, we obtain a sparse noise mask and estimate the subspace robustly after an effective sparse detection step. Then, we embed the estimated sparse noise mask into the fidelity term of the proposed model. Finally, to effectively learn the eigen-images, we jointly use the nonlocal before exploiting the spatial self-similarity of the eigen-images and a weighted group sparsity to promote the group sparsity along the mode-3 of the eigen-images. The objective function of the proposed MixFus model is written as

$$\min_{\mathcal{A}} \left\|\mathbf{M}_{(3)} \odot \left(\widehat{\mathbf{E}}\mathbf{A}_{(3)}\mathbf{BS} - \mathbf{X}_{(3)}\right)\right\|_F^2 + \left\|\mathbf{R}\widehat{\mathbf{E}}\mathbf{A}_{(3)} - \mathbf{Y}_{(3)}\right\|_F^2 + \alpha\phi(\mathcal{A}) + \beta\left\|W \odot D\mathcal{A}\right\|_{2,1} \tag{19}$$

where α and β are regularization parameters. $\phi(\cdot)$ is the regularization of $\mathcal{A}$ to take advantage of the nonlocal self-similarity of the eigen-images.

By introducing four variables, $\mathbf{V}_1$, $\mathcal{V}_2$, $\mathcal{R}$, and $\mathcal{Q}$, we can transform the objective function into a constrained optimization as

$$\begin{aligned} \min_{\mathcal{A},\mathcal{V}_1,\mathcal{V}_2,\mathcal{R},\mathcal{Q}} \; & ||\mathbf{M}_{(3)} \odot \mathbf{V}_1||_F^2 + || \mathbf{R}\widehat{\mathbf{E}}\mathbf{A}_{(3)} - \mathbf{Y}_{(3)}||_F^2 \\ & + \alpha\phi(\mathcal{V}_2) + \beta||W \odot \mathcal{R}||_{2,1} \\ \text{s.t. } & \mathbf{V}_1 = \widehat{\mathbf{E}}\mathbf{A}_{(3)}\mathbf{BS} - \mathbf{X}_{(3)}, \quad \mathcal{V}_2 = \mathcal{A} \\ & D\mathcal{Q} = \mathcal{R}, \quad \mathcal{Q} = \mathcal{A}. \end{aligned} \tag{20}$$

Next, we transformed the constrained problem into the following augmented Lagrangian function:

$$\begin{aligned} & L(\mathcal{A}, \mathbf{V}_1, \mathcal{V}_2, \mathcal{R}, \mathcal{Q}, \mathcal{G}_i) \\ & = ||\mathbf{M}_{(3)} \odot \mathbf{V}_1||_F^2 \\ & + ||\mathbf{R}\widehat{\mathbf{E}}\mathbf{A}_{(3)} - \mathbf{Y}_{(3)}||_F^2 + \alpha\phi(\mathcal{V}_2) + \beta||W \odot \mathcal{R}||_{2,1} \\ & + \mu_1\left\|\mathbf{V}_1 - \widehat{\mathbf{E}}\mathbf{A}_{(3)}\mathbf{BS} + \mathbf{X}_{(3)} + \frac{\mathbf{G}_{1(3)}}{2\mu_1}\right\|_F^2 \\ & + \mu_2\left\|\mathcal{V}_2 - \mathcal{A} + \frac{\mathcal{G}_2}{2\mu_2}\right\|_F^2 + \mu_3\left\|D\mathcal{Q} - \mathcal{R} + \frac{\mathcal{G}_3}{2\mu_3}\right\|_F^2 \\ & + \mu_4\left\|\mathcal{Q} - \mathcal{A} + \frac{\mathcal{G}_4}{2\mu_4}\right\|_F^2 \end{aligned} \tag{21}$$

where $\mu_i (i = 1, 2, 3, 4)$ are the penalty parameters, $\mathcal{G}_i (i = 1, 2, 3, 4)$ are the Lagrange multipliers, and $\mathbf{G}_{1(3)}$ is the mode-3 unfolding of $\mathcal{G}_1$. By referring to the alternative method of multipliers (ADMM), we can split (21) into several small subproblems. Then, we can easily solve each subproblem alternatively.

1) Subproblem of $\mathbf{V}_1$: Subproblem of $\mathbf{V}_1$ can be solved by minimizing $L(\mathcal{A}, \mathbf{V}_1, \mathcal{V}_2, \mathcal{R}, \mathcal{Q}, \mathcal{G}_i)$ with respect to $\mathbf{V}_1$, such that

$$\begin{aligned} \mathcal{V}_1 & \in \arg\min_{\mathcal{V}_1} L(\mathcal{A}, \mathcal{V}_1, \mathcal{V}_2, \mathcal{R}, \mathcal{Q}, \mathcal{G}_i) \\ & = \arg\min_{\mathcal{V}_1} \mu_1\left\|\mathbf{V}_1 - \widehat{\mathbf{E}}\mathbf{A}_{(3)}\mathbf{BS} + \mathbf{X}_{(3)} + \frac{\mathbf{G}_{1(3)}}{2\mu_1}\right\|_F^2 \\ & + ||\mathbf{M}_{(3)} \odot \mathbf{V}_1||_F^2. \end{aligned} \tag{22}$$

Since (22) is strongly convex, by setting (22) with respect to $\mathcal{V}_1$ to zero, we can obtain

$$\mathbf{M}_{(3)} \odot \mathbf{M}_{(3)} \odot \mathbf{V}_1 + \mu_1\mathbf{V}_1 = \mu_1\left(\widehat{\mathbf{E}}\ \mathbf{A}_{(3)}\mathbf{BS} - \mathbf{X}_{(3)} - \frac{\mathbf{G}_{1(3)}}{2\mu_1}\right) \tag{23}$$

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whose vectorizing form is

$$\left(\frac{1}{\mu_1}\text{diag}\left(\mathbf{M}_{(3)}\odot\mathbf{M}_{(3)}\right)+\mathbf{I}\right)\text{vec}(\mathbf{V}_1) = \text{vec}\left(\widehat{\mathbf{E}}\mathbf{A}_{(3)}\mathbf{BS}-\mathbf{X}_{(3)}-\frac{\mathbf{G}_{1(3)}}{2\mu_1}\right) \quad (24)$$

where diag(·) is a function to reshape a tensor into a diagonal matrix whose diagonal elements are the elements of the tensor, vec(·) is a function to vectorize a tensor into a vector, and $\mathbf{I}$ is the unit matrix. Equation (24) can be directly solved as

$$\mathcal{V}_1 = \text{vec}^{-1}\left(\mathbf{T}^{-1}\left(\widehat{\mathbf{E}}\mathbf{A}_{(3)}\mathbf{BS}-\mathbf{X}_{(3)}-\frac{\mathbf{G}_{1(3)}}{2\mu_1}\right)\right) \quad (25)$$

where $\mathbf{T}=((1/\mu_1)\text{diag}(\mathbf{M}_{(3)}\odot\mathbf{M}_{(3)})+\mathbf{I})$, and $\text{vec}^{-1}(\cdot)$ denotes the inverse operation of the vectorization, reshaping a vector to a tensor.

2) Subproblem of $\mathcal{V}_2$: Subproblem of $\mathcal{V}_2$ can be solved by minimizing $L(\mathcal{A},\mathbf{V}_1,\mathcal{V}_2,\mathcal{R},\mathcal{Q},\mathcal{G}_i)$ with respect to $\mathcal{V}_2$ as

$$\mathcal{V}_2 \in \arg\min_{\mathcal{V}_2} L(\mathcal{A},\mathbf{V}_1,\mathcal{V}_2,\mathcal{R},\mathcal{Q},\mathcal{G}_i) = \arg\min_{\mathcal{V}_2}\lambda\phi(\mathcal{V}_2)+\mu_2\left\|\mathcal{V}_2-\mathcal{A}+\frac{\mathcal{G}_2}{2\mu_2}\right\|_F^2. \quad (26)$$

Here, we would like to make use of the recent plug-and-play before regularizing the eigen-images rather than conceiving a hand-crafted regularization $\phi(\mathcal{V}_2)$ [52], [53], [54]. According to the plug-and-play framework, we can solve problem (26) by plugging a denoising operator as follows:

$$\mathcal{V}_2 \leftarrow \text{Denoiser}\left(\mathcal{A}-\frac{\mathcal{G}_2}{2\mu_2},\frac{\lambda}{2\mu_2}\right) \quad (27)$$

where Denoiser(·) is a plugged denoiser to regularize the eigen-images. In this article, we plug the state-of-the-art denoiser, BM3D, to effectively take advantage of the spatial nonlocal self-similarity to denoise each eigen-image.

3) Subproblem of $\mathcal{Q}$: Subproblem of $\mathcal{Q}$ can be solved by minimizing $L(\mathcal{A},\mathbf{V}_1,\mathcal{V}_2,\mathcal{R},\mathcal{Q},\mathcal{G}_i)$ with respect to $\mathcal{Q}$, such that

$$\mathcal{Q} \in \arg\min_{\mathcal{Q}} L(\mathcal{A},\mathbf{V}_1,\mathcal{V}_2,\mathcal{R},\mathcal{Q},\mathcal{G}_i) = \arg\min_{\mathcal{Q}}\mu_3\left\|D\mathcal{Q}-\mathcal{R}+\frac{\mathcal{G}_3}{2\mu_3}\right\|_F^2 + \mu_4\left\|\mathcal{Q}-\mathcal{A}+\frac{\mathcal{G}_4}{2\mu_4}\right\|_F^2. \quad (28)$$

Equation (28) is a least-squared problem, which can be directly calculated by the linear equations system

$$\left(\mu_3 D^T D+\mu_4\mathcal{I}\right)\mathcal{Q} = D^T(\mu_3\mathcal{R}-\mathcal{G}_3)+(\mu_4\mathcal{A}-\mathcal{G}_4) \quad (29)$$

where $\mathcal{I}$ is the unit tensor. As we consider the periodic boundary condition for $\mathcal{Q}$, $D^T D$ possesses block circulant matrices with a circulant block structure. Thus, we can employ the fast Fourier transform (FFT) to efficiently solve (29) as follows [55]:

$$\mathcal{Q} = \text{ifftn}\left(\frac{\mathcal{B}}{\mu_3\mathbf{1}+\mu_4|\text{fftn}(D)|^2}\right) \quad (30)$$

where $\mathcal{B}=D^T(\mu_3\mathcal{R}-\mathcal{G}_3)+(\mu_4\mathcal{A}-\mathcal{G}_4)$, and fftn(·) and ifftn(·) are the fast 3-D Fourier transform and its inverse transform, respectively. The division operation is the elementwise division and $|\cdot|^2$ is the elementwise square.

4) Subproblem of $\mathcal{R}$: Subproblem of $\mathcal{R}$ can be solved by minimizing $L(\mathcal{A},\mathbf{V}_1,\mathcal{V}_2,\mathcal{R},\mathcal{Q},\mathcal{G}_i)$ with respect to $\mathcal{R}$ as

$$\mathcal{R} \in \arg\min_{\mathcal{R}} L(\mathcal{A},\mathbf{V}_1,\mathcal{V}_2,\mathcal{R},\mathcal{Q},\mathcal{G}_i) = \arg\min_{\mathcal{R}}\mu_3\left\|D\mathcal{Q}-\mathcal{R}+\frac{\mathcal{G}_3}{2\mu_3}\right\|_F^2 + \beta||W\odot\mathcal{R}||_{2,1}. \quad (31)$$

Let $\mathcal{C}=D\mathcal{Q}+(\mathcal{G}_3/2\mu_3)$, we can easily obtain the closed-form solution of the tube fibers of $\mathcal{R}$ by using the following calculations:

$$\mathcal{R}(i,j,:) = \text{shrink}_{2,1}\left(\mathcal{C}(i,j,:),|W(i,j,:)|\cdot\frac{\beta}{2\mu_3}\right) \quad (32)$$

where

$$\text{shrink}_{2,1}(\mathbf{x},\xi) = \begin{cases}\dfrac{\|\mathbf{x}\|_2-\xi}{\|\mathbf{x}\|_2}\mathbf{x}, & \text{if } \xi<\|\mathbf{x}\|_2\\ 0, & \text{otherwise}\end{cases} \quad (33)$$

$$W_x(i,j,:) = \frac{1}{\left\|\left(D_x\mathcal{Q}+\frac{\mathcal{G}_3}{2\mu_3}\right)(i,j,:)\right\|_2+\epsilon} \quad (34)$$

$$W_y(i,j,:) = \frac{1}{\left\|\left(D_y\mathcal{Q}+\frac{\mathcal{G}_3}{2\mu_3}\right)(i,j,:)\right\|_2+\epsilon} \quad (35)$$

and ϵ is a small constant to avoid singularities.

5) Subproblem of $\mathcal{A}$: Subproblem of $\mathcal{A}$ can be solved by minimizing $L(\mathcal{A},\mathbf{V}_1,\mathcal{V}_2,\mathcal{R},\mathcal{Q},\mathcal{G}_i)$ with respect to $\mathcal{A}$ as

$$\begin{aligned}\mathcal{A} \in \arg\min_{\mathcal{A}} & L(\mathcal{A},\mathbf{V}_1,\mathcal{V}_2,\mathcal{R},\mathcal{Q},\mathcal{G}_i)\\ = & ||\mathbf{R}\widehat{\mathbf{E}}\mathbf{A}_{(3)}-\mathbf{Y}_{(3)}||_F^2\\ & +\mu_1\left\|\mathbf{V}_1-\widehat{\mathbf{E}}\mathbf{A}_{(3)}\mathbf{BS}+\mathbf{X}_{(3)}+\frac{\mathbf{G}_{1(3)}}{2\mu_1}\right\|_F^2\\ & +\mu_2\left\|\mathcal{V}_2-\mathcal{A}+\frac{\mathcal{G}_2}{2\mu_2}\right\|_F^2+\mu_4\left\|\mathcal{Q}-\mathcal{A}+\frac{\mathcal{G}_4}{2\mu_4}\right\|_F^2.\end{aligned} \quad (36)$$

The solver of $\mathbf{A}$ can be obtained by setting the derivative of (36) with respect to $\mathbf{A}$ to zero due to strongly convex of (36). By taking the $\mathbf{E}^T\mathbf{E}=\mathbf{I}$ into consideration, we can obtain the following Sylvester equation:

$$\mathbf{H}_1\mathbf{A}+\mathbf{A}\mathbf{H}_2 = \mathbf{H}_3 \quad (37)$$

where

$$\begin{aligned}\mathbf{H}_1 &= \frac{\alpha}{\mu_1}\left(\mathbf{R}\widehat{\mathbf{E}}\right)^T\mathbf{R}\widehat{\mathbf{E}}+\frac{\mu_2}{\mu_1}\mathbf{I}+\frac{\mu_4}{\mu_1}\mathbf{I}\\ \mathbf{H}_2 &= (\mathbf{BS})(\mathbf{BS})^T\\ \mathbf{H}_3 &= \widehat{\mathbf{E}}^T\left(\mathbf{V}_1+\mathbf{X}_{(3)}+\frac{\mathbf{G}_{1(3)}}{2\mu_1}\right)(\mathbf{BS})^T\\ &\quad+\frac{\alpha}{\mu_1}\left(\mathbf{R}\widehat{\mathbf{E}}\right)^T\mathbf{Y}_{(3)}+\frac{\mu_2}{\mu_1}\left(\mathbf{V}_{2(3)}+\frac{\mathbf{G}_{2(3)}}{2\mu_2}\right)\\ &\quad+\frac{\mu_4}{\mu_1}\left(\mathbf{V}_{4(3)}+\frac{\mathbf{G}_{4(3)}}{2\mu_4}\right)\end{aligned} \quad (38)$$

which can be solved analytically and efficiently.

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Algorithm 1 Optimization of the Objective Function for the Proposed **MixFus** Fusion Method

Input: The LR-HSI $\mathcal{X}$, HR-MSI $\mathcal{Y}$, regularization parameters α and β.
Initialize: μ_i, $\mathbf{V}_1$, $\mathcal{V}_2$, $\mathcal{Q}$, $\mathcal{R}$, $\mathcal{G}_i$=0.
Generate sparse noise mask $\mathcal{M}$ and estimate the subspace $\widehat{\mathbf{E}}$.
repeat
  Calculate $\mathbf{V}_1$ via (25)
  Calculate $\mathcal{V}_2$ via (27)
  Calculate $\mathcal{Q}$ via (30)
  Calculate $\mathcal{R}$ via (32)
  Calculate $\mathcal{A}$ via (37)
  Update $\mathcal{G}_i$ and μ_2 via (39)
until stopping criterion is satisfied.
Output: The restored HSI $\mathcal{X} = \mathcal{A}\times_3\widehat{\mathbf{E}}$

6) Updating Multipliers: The Lagrange multipliers $\mathcal{G}_i (i = 1, 2, 3, 4)$ and penalty parameter μ_2 are updated as follows:

$$\begin{cases} \mathbf{G}_{1(3)} = \mathbf{G}_{1(3)} + 2\mu_1\left(\mathbf{V}_1 - \widehat{\mathbf{E}}\mathbf{A}_{(3)}\mathbf{B}\mathbf{S} + \mathbf{X}_{(3)}\right) \\ \mathcal{G}_2 = \mathcal{G}_2 + 2\mu_2(\mathcal{V}_2 - \mathcal{A}) \\ \mathcal{G}_3 = \mathcal{G}_3 + 2\mu_3(D\mathcal{Q} - \mathcal{R}) \\ \mathcal{G}_4 = \mathcal{G}_4 + 2\mu_4(\mathcal{Q}_4 - \mathcal{A}) \\ \mu_2 = \min(\mu, \rho\mu_2) \end{cases} \tag{39}$$

where μ and $\rho > 1$ denote constant values.

To better understand the proposed method, we present the procedures to optimize the objective function (19) in Algorithm 1. In Algorithm 1, the number of iterations is set to 30 for the sake of saving time.

IV. Experiments

In this section, we conducted extensive experiments on three reduced-resolution datasets and a full-resolution dataset to fully evaluate the fusion performance of the proposed method in comparison with seven state-of-the-art competitors.

A. Simulation of Observed Images

Three commonly used HSI datasets, that is, Pavia University,[1] Washington DC Mall,[2] and Hyperion Cuprite,[3] is used for the simulation of the observed images at reduced resolution. The Pavia University used has a size of 256 × 256 × 93 [see Fig. 3(a)], the Washington DC Mall used has a size of 256 × 256 × 191 [see Fig. 3(b)], and the Hyperion Cuprite used has a size of 256 × 256 × 188 [see Fig. 3(c)]. We followed Wald's protocol to simulate the observed LR-HSIs and HR-MSIs [19], [56]. Specifically, we first applied an isotropic Gaussian PSF [19] to the HR-HSIs to obtain the blurred HSIs. Then, we downsampled the blurred HSIs with downsampling ratios equal to 8, 4, and 2 for Pavia University, Washington DC Mall, and Hyperion Cuprite, respectively,

[1]https://rslab.ut.ac.ir/data
[2]http://lesun.weebly.com/hyperspectral-dataset.html
[3]https://aviris.jpl.nasa.gov/data/free_data.html

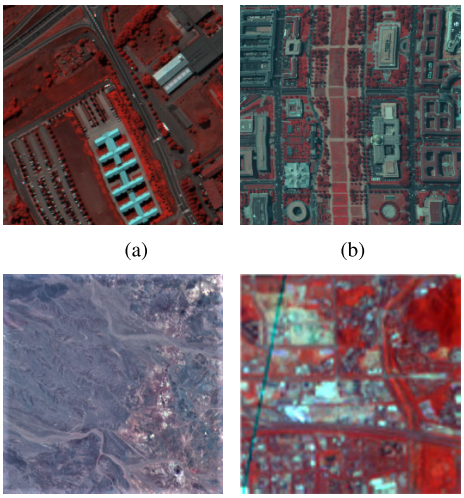


Fig. 3. Hyperspectral datasets used in the experiments. (a) Pavia University. (b) Washington DC Mall. (c) Hyperion Cuprite. (d) Hyperion Shenzhen.

to evaluate the fusion performance with different downsampling ratios. The HR-MSI for Pavia University was generated using the IKONOS-like reflectance spectral response [57]. The HR-MSIs for Washington DC Mall and Hyperion Cuprite were generated using the response function of the Sentinel-2A and Landsat 8 instruments, respectively [58]. To simulate noisy observations, white Gaussian noise (WGN) was added to the HR-MSI to generate a noisy observed HR-MSI with a signal-to-noise ratio (SNR) equal to 40 dB. For the LR-HSIs, we simulated the noisy observations by adding four different types of noise to the LR-HSIs, including nonindependent and identically distributed (non-i.i.d.) Gaussian, "Salt & Pepper" noise, and stripes noises, as described in Cases 1–4.

Case 1 (Gaussian Non-i.i.d. Noise): Non-i.i.d. Gaussian noise is added to the LR-HSI to generate a noisy observed LR-HSI with its SNR equal to 35 dB.

Case 2 (Gaussian Noise + Stripes): Gaussian noise (depicted in Case 1) and oblique stripe noise. The stripes randomly affect nearly 10% bands of the images. On each band, there are approximately 5–15 stripes present.

Case 3 (Gaussian Noise + "Salt & Pepper" Noise): Gaussian noise (depicted in Case 1) and "Salt & Pepper" noise. The noise density of the "Salt & Pepper" noise is 1%.

Case 4 (Gaussian Noise + Stripes + "Salt & Pepper" Noise): Gaussian noise (depicted in Case 1), random oblique stripes (depicted in Case 2), and "Salt & Pepper" noise (depicted in Case 3).

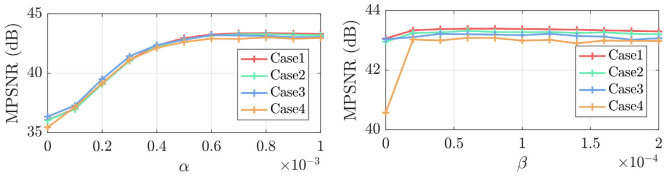
B. Competitors and Evaluation Indices

Seven state-of-the-art fusion techniques are employed as comparative methods to demonstrate the superiority of the proposed approach. They are GLPHS [19], CNMF [20], HySure [21], LTMR [22], CSTF [59], NLSTF [28], and LRTA [34]. Among them, GLPHS is a representative pansharpening method extended for fusing HSIs and MSIs. CNMF and HySure are benchmarks of the unmixing-based fusion methods. LTMR is a state-of-the-art subspace representation-based

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TABLE I
MAIN PARAMETERS FOR COMPARATIVE METHODS

Method	The values of the main parameters
GLPHS	-
CNMF	$K = 30$
HySure	$\lambda_\phi = 5e-4$, $\lambda_m = 1$
LTMR	$K = 201$, $\lambda = 1e-3$
CSTF	$\lambda = 1e-5$, $\beta = 0.1$
NLSTF	$\lambda = 1e-6$, $\lambda_1 = \lambda_2 = 1e-5$, $K = 151$
LRTA	$\omega_1 = \omega_2 = 1$, $\omega_3 = 100$



(a) (b)

Fig. 4. MPSNR values of the proposed method as a function of the parameter (a) α and (b) β on the Pavia University data.

fusion method. CSTF, NLSTF, and LRTA are state-of-the-art tensor-based fusion methods.

The parameter settings for all the comparative methods are obtained from their corresponding papers or original source codes, the specific values for the main regularization parameters for the comparative methods can be found in Table I. All experiments were implemented in MATLAB (R2016a) on Windows 10 with an Intel Core i7-7700HQ 2.8-GHz processor.

To perform quantitative evaluation of the denoising fusion results, we employ five metrics, namely, the mean peak SNR (MPSNR), the root mean squared error (RMSE), the spectral angle mapper (SAM) [60], the relative dimensionless global error in synthesis (ERGAS) [61], and the universal image quality index (UIQI) [62], to analyze the experimental outcomes for the reduced resolution datasets. Due to the absence of a reference image for the full-resolution dataset, we adopted a modified quality with no reference (mQNR) metric [63] to quantitatively assess the fusion performance at full resolution. The maximum value of the mQNR index is 1, which is achieved when both spectral and spatial distortions are absent.

C. Analysis of Parameters

Two regularization parameters, α and β, need to be tuned in our proposed method. To determine appropriate values for the two parameters, we present the fusion performance under different values of α and β in Fig. 4, using the Pavia University data as an example. Parameter α controls the degree to which nonlocal self-similarity is employed for the regularization of eigen-images. As shown in Fig. 4(a), we can easily observe that the MPSNR values of the fusion results increase with α at first and then become stable after reaching 0.6×10^{-3}. During this process, the MPSNR increases by approximately 7 dB, which is a significant improvement. Thus, we set the value of α to be 0.8×10^{-3}.

Fig. 4(b) presents the fusion results with different values of parameter β on the Pavia University dataset. Parameter β controls the group sparsity among eigen-images in spatial differences. It can be observed that the MPSNR of the experimental results increases when β is smaller than 0.5×10^{-4} and then becomes relatively stable with the increase of β. In this article, we set β to 1.2×10^{-4} in our experiments.

D. Experimental Results for Reduced-Resolution Images

1) Experimental Results Under Four Noise Cases: Table II presents the specific numerical values of each metric under different simulation scenarios for the three datasets, respectively. The best results are highlighted in bold. It can be observed that most of the methods achieve relatively satisfactory quantitative results when only Gaussian noise is present, as depicted in Case 1, except that GLPHS and CNMF can only obtain poor results at Pavia University. Among all the methods, the proposed method produces the best results for all three datasets. The exploitation of the nonlocal self-similarity and group sparsity characteristics along the mode-3 of the eigen-images enables the proposed method to regularize eigen-images effectively, producing high-quality fusion results. Among the competitors, LRTA, NLSTF, and CSTF produce results very close to our method for Pavia University, Washington DC Mall, and Curpite data, respectively, in the presence of Gaussian noise. However, in Cases 2–4, the results obtained by the seven comparative methods were relatively poor, while our proposed method still robustly obtained a high-quality result. It can be inferred that the presence of mixed noise in the image can significantly undermine the effectiveness of MSI and HSI fusion. Since the impact of mixed noise is not taken into consideration by the competitors, the poor fusion results of the competitors are predictable. Nevertheless, our fusion method estimates the subspace after an effective sparse noise detection step, which can effectively suppress the impact of mixed noise on the estimation of the subspace, thus robustly obtaining high-quality fusion results. The running time of each method on three datasets is also presented in Table II, from which we can see that the proposed method achieves the most effective fusion performance at the sacrifice of the running time.

In Fig. 5, we plot the band-wise PSNR of the three datasets under different scenarios to quantitatively evaluate the performance of the proposed fusion method on each band. It is evident from Fig. 5 that, in the case of simple Gaussian noise as in Case 1, LTMR, CSTF, NLSTF, LRTA, and our proposed method all achieve comparable good fusion results on each band, with our method slightly outperforming the others. However, when sparse noise is present, that is, from the second to the fourth column of Fig. 5, the other seven comparative methods are severely affected by the mixed noise, which leads to compromised fusion results. By observing the second column, we can infer that the fusion results of a band will be severely degraded when stripe noise appears in that band for the competitors. The results in the third column, corresponding to mixed noise in Case 3, similarly demonstrate the adverse impact of sparse noise on the process of image fusion for the competitors. In contrast, our proposed method achieves the overall best on all the bands under various scenarios, no matter in the most complicated Case 4 or the relatively simple Case 1 with only Gaussian noise, which

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TABLE II
QUANTITATIVE INDICES OF THE PROPOSED METHOD AND COMPARED METHODS ON THREE DATASETS WITH FOUR TYPES OF NOISES

Data	Cases	Indices	GLPHS	CNMF	HySure	LTMR	CSTF	NLSTF	LRTA	Proposed
Pavia University	Case 1	MPSNR↑	27.09	33.63	41.87	41.01	42.89	42.23	43.00	**43.37**
		RMSE↓	11.33	5.45	2.20	2.82	1.99	2.12	2.03	**1.96**
		SAM↓	6.42	3.79	2.15	2.52	2.03	2.06	2.00	**1.82**
		ERGAS↓	3.37	1.55	0.62	0.73	0.54	0.60	0.54	**0.52**
		UIQI↓	0.88	0.96	0.99	0.99	0.99	0.99	0.99	**0.99**
		Time(s)↓	**6**	13	107	63	71	20	42	184
	Case 2	MPSNR↑	25.49	30.09	33.43	32.24	33.23	34.70	35.73	**43.28**
		RMSE↓	19.91	21.88	19.51	21.66	20.23	14.75	16.87	**1.98**
		SAM↓	15.80	20.58	16.18	19.48	20.48	15.44	13.67	**1.83**
		ERGAS↓	4.42	5.08	4.22	4.76	5.01	2.69	3.84	**0.53**
		UIQI↓	0.82	0.89	0.93	0.91	0.89	0.93	0.93	**0.99**
		Time(s)↓	**6**	12	107	60	69	24	40	183
	Case 3	MPSNR↑	26.07	30.63	29.55	34.49	33.53	30.87	27.68	**43.22**
		RMSE↓	12.81	9.99	10.27	6.86	8.26	7.67	10.88	**2.14**
		SAM↓	8.26	9.93	10.61	6.65	8.97	8.20	12.13	**1.92**
		ERGAS↓	3.76	3.67	3.38	2.27	2.79	2.31	4.66	**0.64**
		UIQI↓	0.83	0.89	0.86	0.94	0.92	0.92	0.87	**0.99**
		Time(s)↓	**7**	10	107	59	70	24	38	186
	Case 4	MPSNR↑	24.83	30.20	27.92	31.22	31.76	29.27	26.16	**42.85**
		RMSE↓	20.30	20.19	21.53	22.10	21.31	15.14	19.80	**2.16**
		SAM↓	16.07	18.08	20.47	20.53	22.25	15.92	19.65	**2.01**
		ERGAS↓	4.55	4.72	5.24	4.92	5.30	3.26	6.04	**0.57**
		UIQI↓	0.78	0.87	0.83	0.89	0.88	0.88	0.82	**0.99**
		Time(s)↓	**7**	11	107	58	70	24	52	179
Washington DC Mall	Case 1	MPSNR↑	40.03	45.41	46.17	47.61	49.16	49.59	43.13	**50.44**
		RMSE↓	5.28	2.01	2.19	1.96	1.22	1.05	3.79	**1.04**
		SAM↓	4.67	1.95	2.61	2.46	1.50	1.24	4.03	**1.23**
		ERGAS↓	5.65	4.04	3.49	4.29	2.80	5.76	16.27	**2.57**
		UIQI↓	0.92	0.96	0.97	0.96	0.98	0.96	0.85	**0.99**
		Time(s)↓	30	**27**	107	59	80	35	67	177
	Case 2	MPSNR↑	37.14	40.27	40.62	38.46	40.01	43.04	38.59	**50.41**
		RMSE↓	14.73	14.46	19.27	19.59	16.60	15.73	18.92	**1.04**
		SAM↓	16.07	18.04	19.44	25.63	22.93	21.74	18.22	**1.23**
		ERGAS↓	13.92	17.32	23.58	24.73	235.54	62.34	28.55	**2.61**
		UIQI↓	0.85	0.87	0.88	0.84	0.83	0.87	0.78	**0.99**
		Time(s)↓	31	**18**	104	60	83	35	73	172
	Case 3	MPSNR↑	32.78	33.31	31.52	35.86	36.20	30.25	24.49	**48.98**
		RMSE↓	6.66	13.55	8.52	5.70	5.21	8.21	15.45	**2.03**
		SAM↓	7.57	26.80	12.16	8.24	7.99	12.23	23.06	**1.78**
		ERGAS↓	17.74	77.69	93.11	70.22	134.25	81.77	189.57	**35.31**
		UIQI↓	0.63	0.64	0.56	0.66	0.66	0.54	0.53	**0.97**
		Time(s)↓	31	**14**	100	60	79	38	67	166
	Case 4	MPSNR↑	31.04	37.20	32.92	35.41	36.87	30.35	23.33	**48.92**
		RMSE↓	14.81	15.87	20.33	19.88	16.72	15.49	24.03	**6.38**
		SAM↓	17.10	18.08	22.78	26.20	23.62	21.53	30.49	**5.19**
		ERGAS↓	19.77	16.82	47.45	27.18	104.71	134.06	157.90	**7.43**
		UIQI↓	0.61	0.73	0.62	0.71	0.72	0.56	0.50	**0.96**
		Time(s)↓	32	**19**	104	61	75	37	69	166
Hyperion Cuprite	Case 1	MPSNR↑	40.89	42.39	38.08	41.15	43.20	42.13	39.06	**45.12**
		RMSE↓	2.58	2.41	3.70	3.74	2.30	2.76	3.00	**1.95**
		SAM↓	0.87	0.86	1.12	1.35	0.84	1.02	1.24	**0.75**
		ERGAS↓	0.93	0.85	1.33	1.36	0.80	0.96	1.09	**0.67**
		UIQI↓	0.97	0.98	0.96	0.95	0.97	0.97	0.95	**0.98**
		Time(s)↓	44	127	112	59	89	**43**	105	223
	Case 2	MPSNR↑	39.68	38.70	35.94	38.14	38.23	39.24	37.08	**45.13**
		RMSE↓	4.23	7.58	8.08	8.06	6.63	7.90	7.82	**1.95**
		SAM↓	1.53	2.79	2.77	3.11	2.44	3.10	2.75	**0.75**
		ERGAS↓	1.51	2.81	2.94	2.93	2.39	2.79	2.82	**0.68**
		UIQI↓	0.95	0.93	0.92	0.92	0.93	0.92	0.91	**0.98**
		Time(s)↓	**42**	120	110	57	89	43	105	222
	Case 3	MPSNR↑	37.23	35.17	30.25	36.26	35.30	29.55	25.59	**44.64**
		RMSE↓	3.78	5.09	9.35	4.83	5.40	9.22	13.43	**2.07**
		SAM↓	1.37	1.95	3.44	1.76	1.99	3.50	7.64	**0.78**
		ERGAS↓	1.37	1.88	3.51	1.76	1.99	3.43	7.63	**0.72**
		UIQI↓	0.94	0.90	0.76	0.91	0.89	0.82	0.48	**0.98**
		Time(s)↓	43	112	111	58	90	**42**	107	231
	Case 4	MPSNR↑	36.37	37.52	31.71	35.65	35.56	29.91	25.03	**44.79**
		RMSE↓	4.75	8.11	10.58	9.52	7.89	9.26	15.19	**2.03**
		SAM↓	1.81	3.09	4.28	3.77	3.02	3.62	8.35	**0.77**
		ERGAS↓	1.70	2.97	4.20	3.64	3.01	3.39	8.05	**0.70**
		UIQI↓	0.92	0.91	0.80	0.89	0.89	0.82	0.47	**0.98**
		Time(s)↓	43	91	109	58	91	**41**	107	230

indicates the robustness of the proposed mixed noise-oriented fusion method.

Furthermore, to provide visual comparisons, we present the fused images and the corresponding error maps of one band for the three datasets. Specifically, Fig. 6 shows the fusion result of band 80 in the Pavia University dataset, Fig. 7 shows the fusion result of band 80 in the DC dataset, and Fig. 8 shows the fusion result of band 139 in the Cuprite dataset. When the HSI contains only Gaussian noise, all methods achieve good fusion effects. When the "Salt & Pepper" noise exists, as depicted in Case 3, all methods except for LRTA can remove the "Salt & Pepper" noise to some extent. In the presence of stripe noise as seen in Cases 2 and 4, although NLSTF can remove the stripe noise to a certain extent, the overall image fusion performance is still unsatisfactory. The remaining comparison methods are unable to perfectly eliminate the stripe noise. In contrast, our proposed method achieves excellent fusion results in all four cases of noise, indicating that our method is more applicable and robust when dealing with noisy data.

2) Experimental Results of Different SNR Values: To test the robustness of the proposed method under different

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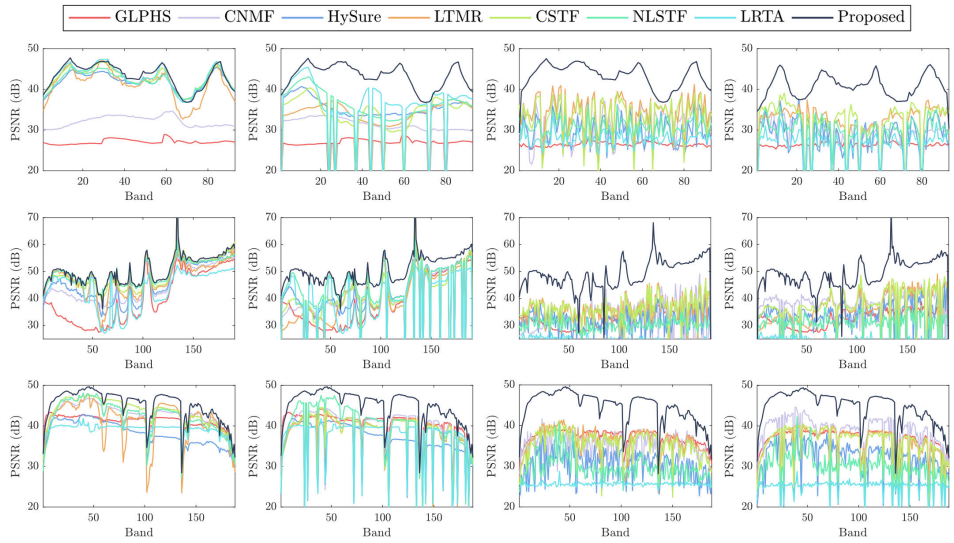


Fig. 5. Band-wise PSNR values for the three datasets. (First row to third row) Pavia University, Washington DC, and Hyperion Cuprite, respectively. (First column to fourth column) Cases 1–4, respectively.

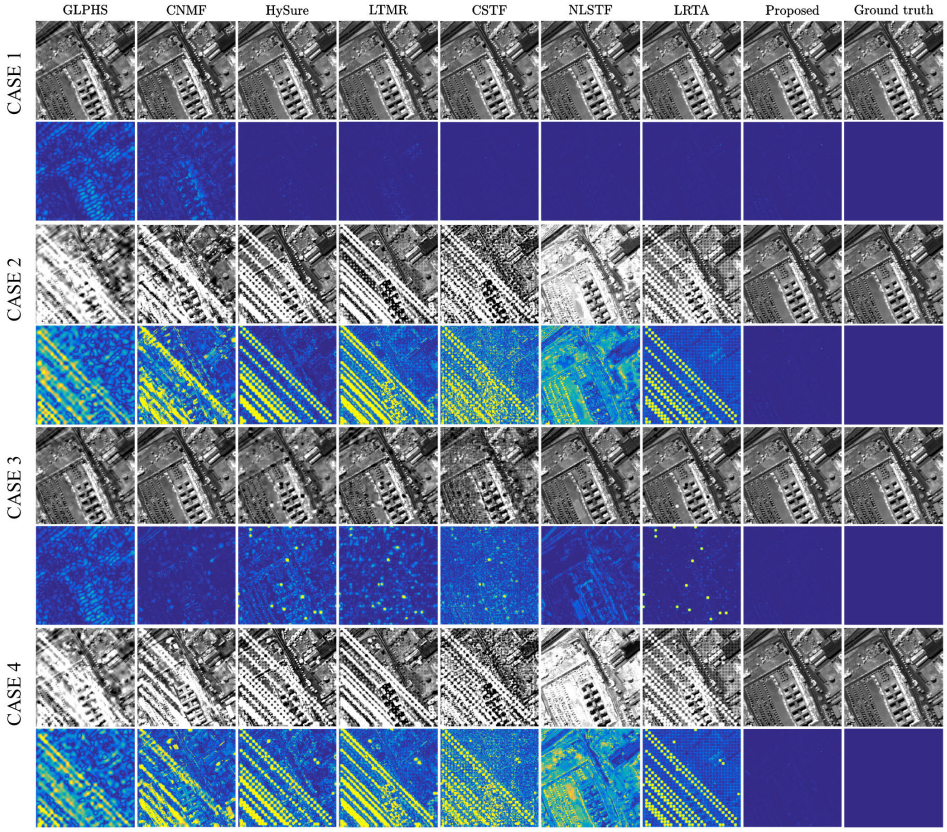


Fig. 6. 80th band of the fusion results of the Pavia University data in four cases.

intensities of noise, we conducted a series of fusion experiments by fusing the HR-MSIs with the LR-HSIs of different SNRs using all the tested methods. The noisy LR-HSIs with different SNRs are obtained by adding non-i.i.d. Gaussian noise with varying intensities, as depicted in Case 1, on the clean LR-HSIs. We also conducted fusion experiments on the clean LR-HSIs without adding any noise, that is, SNR $= \infty$, to evaluate the proposed method with the comparisons fully. Fig. 9 displays the experimental outcomes for three datasets, that is, Pavia University, Washington DC Mall, and Curpite data. It is evident from Fig. 9 that our proposed method surpasses other comparison methods when the SNR equals 30, 35, and 40 dB. Additionally, our approach achieves excellent fusion performance on fusion noise-free images, that is, SNR $= \infty$, demonstrating the robustness of the proposed method in fusing images.

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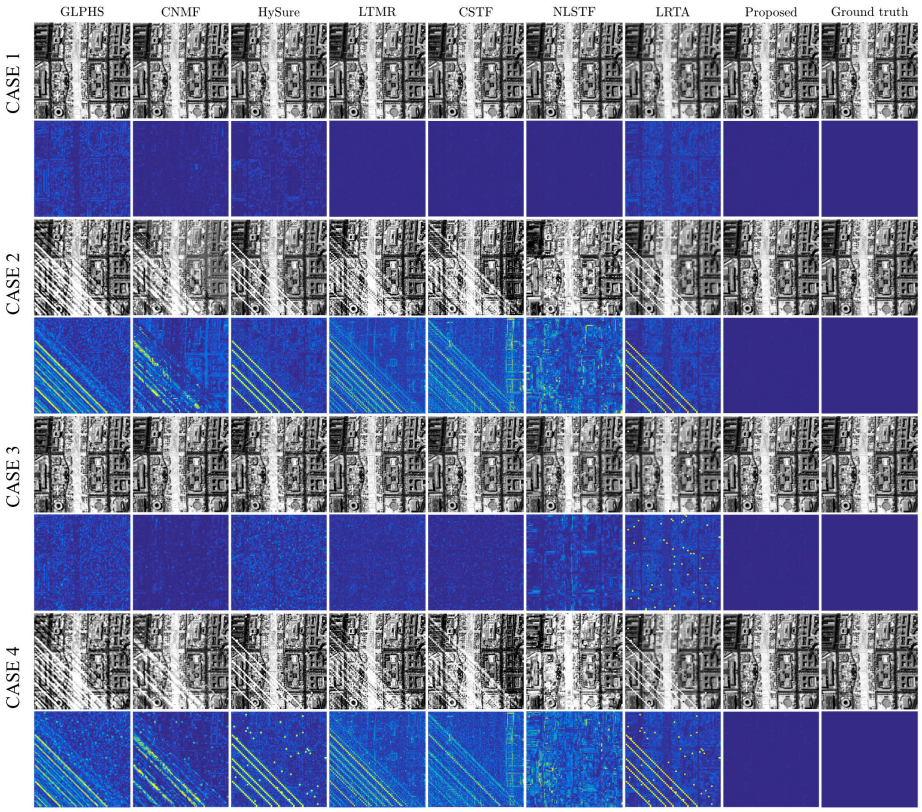


Fig. 7. 80th band of the fusion results of the Washington DC Mall data in four cases.

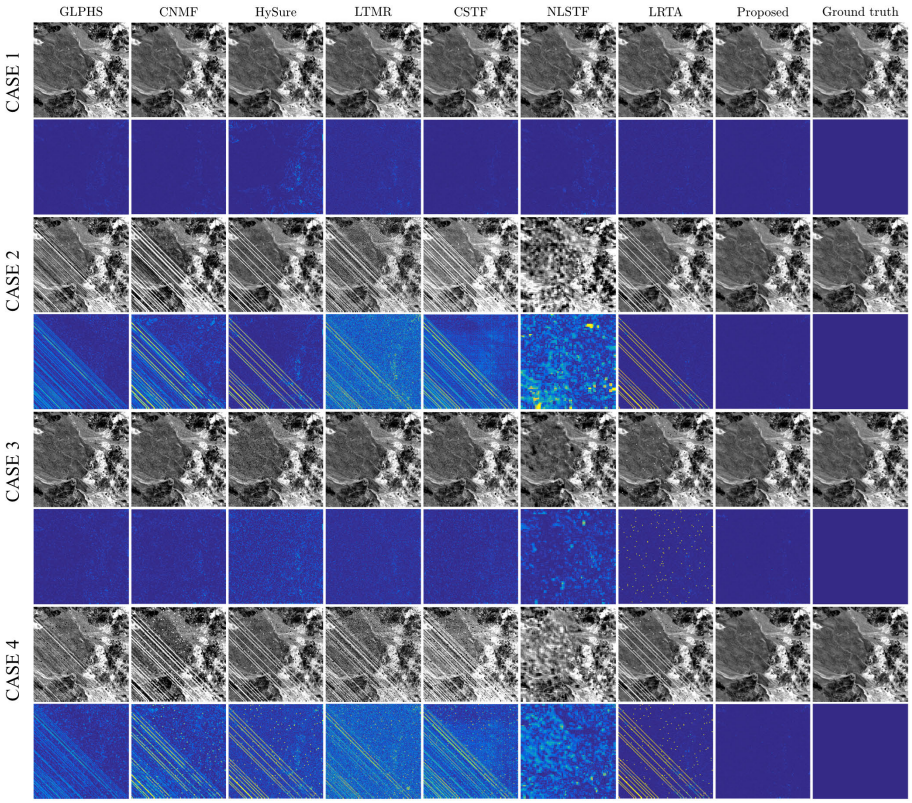


Fig. 8. 139th band of the fusion results of the Hyperion Cuprite data in four cases.

3) Experimental Results of Denoising Before Fusion: Considering that the seven comparison methods only involve the fusion processes without taking into account the influence of noise, for further comparison, a denoising step is carried out before image fusion for the comparison methods. The experiments were performed on Case 4 for the three

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TABLE III
MPSNR VALUES OF THE EXPERIMENTAL RESULTS OF DENOISING BEFORE FUSION ON THREE DATASETS

Data	GLPHS	CNMF	HySure	LTMR	CSTF	NLSTF	LRTA	Proposed
Pavia University	27.62	32.90	38.52	35.03	32.16	41.27	37.44	**43.07**
Washington DC Mall	30.99	37.66	35.66	39.82	31.31	43.24	27.67	**44.53**
Hyperion Cuprite	38.68	38.56	35.36	34.54	38.52	38.00	39.52	**41.60**

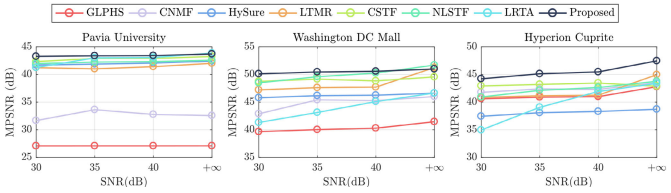


Fig. 9. Fusion results of the three datasets with varying SNR values.

TABLE IV
MPSNR VALUES OF THE ABLATION STUDY ON THE PAVIA UNIVERSITY DATA

Cases	Subspace estimated from noisy images	$\alpha = 0$	$\beta = 0$	Proposed
Case 1	43.37	36.06	43.06	**43.37**
Case 2	34.27	36.03	42.96	**43.28**
Case 3	39.39	36.34	43.04	**43.22**
Case 4	33.16	35.44	40.57	**42.85**

datasets. Specifically, before image fusion, we first denoised the LR-HSIs with mixed noise using the E-3DTV [64] method. Then, we fused the denoised LR-HSIs with HR-MSIs using the comparison methods. For the proposed method, the noisy LR-HSIs and HR-MSIs were directly fused to generate fused images without any denoising steps. Fig. 10 displays the visual results of the three datasets, while Table III shows the corresponding PSNR numerical results. It is evident that our proposed method still outperforms the other seven competing methods. This proves that our method can robustly achieve better performance when dealing with the fusion problem in the presence of noise.

E. Ablation Study

We conducted a series of ablation experiments to demonstrate the effectiveness of the main contributions of the proposed method using Pavia University data. The results of the ablation experiments are presented in Table IV, where the best MPSNR values are highlighted in bold for convenience. First, we directly perform subspace estimation, that is, SVD, on the noisy HSI data without using the proposed subspace estimation method. The results are presented in the first column of Table IV. It can be clearly observed that when the HSI only contains Gaussian noise as depicted in Case 1, the fusion result obtained is identical to that of the proposed method. This is because, under the Gaussian noise, the subspace obtained from the ablation experiment is the same as that obtained by the proposed method. However, when the image is affected by mixed noise, that is, the scenarios depicted in Cases 2–4, the MPSNR values of the ablation experiments decrease significantly compared with the proposed method. This proves that subspace estimation directly obtained from the noisy image cannot accurately estimate the subspace when mixed noise is present, thus leading to worse fusion results. Our method estimates the subspace after the sparse noise detection and thus can obtain robust fusion performance in these circumstances.

The second column in Table IV shows the results of $\alpha = 0$, which means that the BM3D denoiser is not workable. It can be easily observed that the results obtained from the ablation experiment without regularization by the inserted denoiser are far inferior to the proposed method, which proves the effectiveness of spatial self-similarity in regularizing eigen-images.

Finally, we conducted fusion experiments without imposing weighted group sparsity on the eigen-images, that is, $\beta = 0$. The experimental results are given in the penultimate column of Table IV. In the four different cases, the results are all worse than those of the proposed method, which indicates that the introduction of weighted group sparsity to regularize the eigen-image can further improve the fusion performance.

As demonstrated above, it is readily apparent that the utilization of the BM3D denoiser to regularize the eigen-images constitutes a pivotal component of our proposed method, as reflected by the ablation experiments we conducted. When confronted with intricate noise patterns, implementing a subspace estimation technique can substantially enhance the fusion effect. Furthermore, the incorporation of weighted group sparsity can serve to improve the fusion performance further.

F. Experimental Results for Real Full-Resolution Images

To evaluate the effectiveness of the proposed method, we conducted a full-resolution fusion experiment using a pair of real MSIs and HSIs containing actual stripe noise in the HSI. The HSI, named Hyperion Shenzhen, was captured in Shenzhen, Guangdong Province, China, by the Hyperion sensor mounted on the Earth Observing-1 satellite in 2015. It comprises 242 bands with a spatial resolution of 30 m. After excluding some defective bands, a $100 \times 100 \times 198$ subimage is cropped as the experimental dataset. In the HSI, explicit striping artifacts exist in specific bands, namely bands 61, 109, 119, 136, and 172. We present the false-color image composed of bands 61, 13, and 5, in Fig. 3(d), in which a stripe can be obviously observed. Simultaneously, we acquired the multispectral image data of the same area from the Sentinel-2 satellite, with a spatial resolution of 10 m.

Fig. 11 displays pseudocolor images composed of three bands (61, 13, and 5) to showcase the fusion results of each method. The zoomed-in images in the red boxes at the lower right corner of the images show a local enlargement

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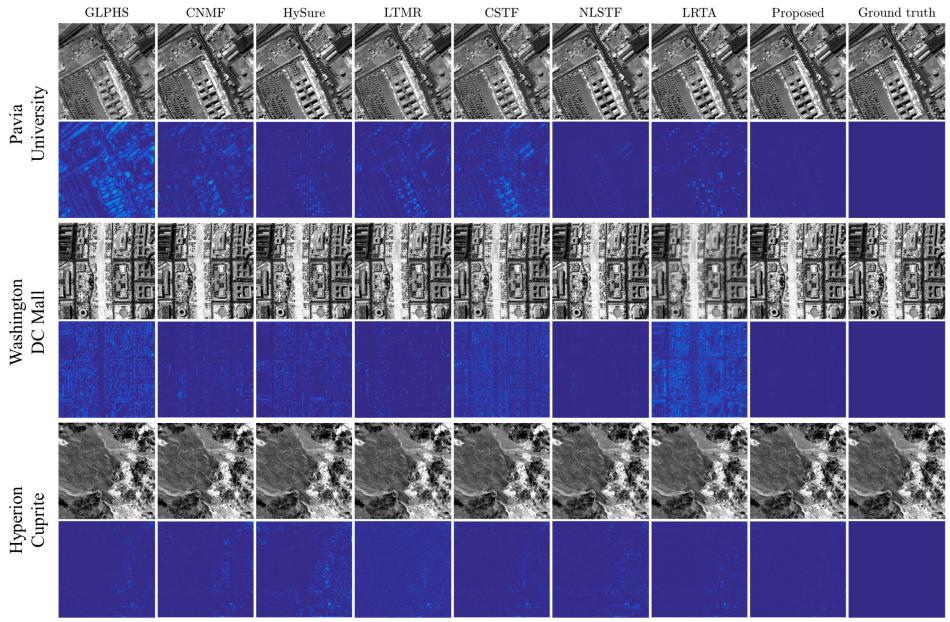


Fig. 10. Experimental results of denoising before fusion on the Pavia University data (band 80), Washington DC Mall data (band 80), and Hyperion Cuprite data (band 139).

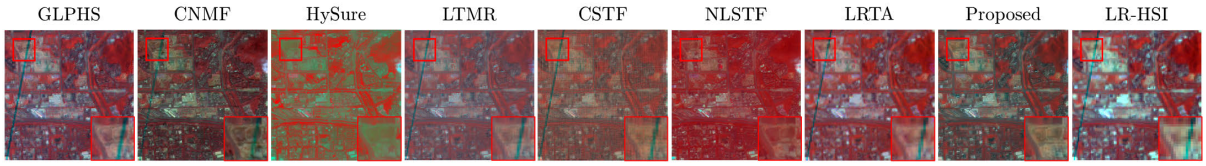


Fig. 11. Fusion results of the Hyperion Shenzhen data. The presented image is composed of bands 61, 13, and 5.

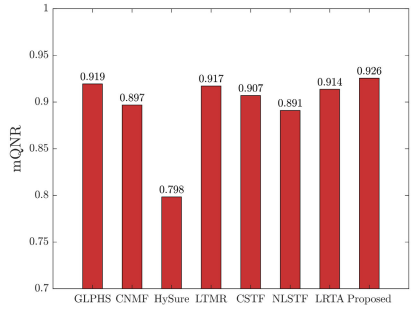


Fig. 12. mQNR values obtained with all tested methods on the Hyperion Shenzhen data.

of the areas marked by the red rectangular boxes in the left upper corner of the images. We can clearly observe that all the methods can improve spatial resolution to some degree. However, GLPHS, CNMF, LTMR, CSTF, and LRTA fail to remove the striping noise in Band 61 of the HSI. Although HySure and NLSTF are capable of noise removal, their fusion results suggest a significant spectral distortion from the LR-HSI. Our proposed method, on the other hand, can obtain a more realistic HR-HSI by improving the spatial resolution and eliminating the stripe noise in the LR-HSI, which shows more potential for practical applications. Meanwhile, we have computed the mQNR for each image and presented the values in Fig. 12. Remarkably, our approach has yielded the best mQNR values in HR-HSI reconstruction, which verifies the effectiveness of the proposed mixed noise-oriented fusion method on fusing real MSIs and HSIs containing actual noise.

V. Conclusion

This article proposes a novel method for the fusion of MSIs and HSIs by estimating the subspace robustly and regularized representation coefficients effectively to mitigate the effects of mixed noise in HSIs. First, our proposed method applies a GMM to detect sparse noise in the image and then robustly estimate the subspace after the sparse noise detection. To exploit the spatial self-similarity characteristics in the eigen-images, we introduce a nonlocal image denoiser to regularize the representation coefficients of the subspace. Furthermore, we propose to impose a weighted group sparse regularizer on eigen-images to take advantage of the group sparsity of spatial difference images among eigen-images, thereby further improving the quality of the fused images. To evaluate the effectiveness of our proposed method, we compare it with seven state-of-the-art approaches using three simulated HSIs and a pair of real HSIs and MSIs. The experimental results demonstrate that our proposed fusion method can achieve better fusion performance in comparison with the competitors, in particular when LR-HSIs are subjected to mixed noise.

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